

Experiment 7

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Semester: IInd
Subject Name: Technical Training

UID: 25MCC20062
Section/Group: 25MCC-101/A
Date of Performance: 31/03/2026
Subject Code: 25CAP-652

Title: Joins in SQL Databases.

1. Aim:

- a. Implementation of joins in PostgreSQL (inner join, left join, right join, self-join and cross join)

2. Software Requirements:

- a. PostgreSQL (Database Server)
- b. pgAdmin
- c. OS: Windows

3. Objective:

- a. Apply joins to a real-world database schema (e.g., Students, Courses, Enrollments, Departments).

4. Procedure of the Practical:

- a. Start the system.
- b. Open pgAdmin.
- c. Create and select the database in which you want to perform the experiment.
- d. Establish connection to the database using Alt+Shift+Q.
- e. Run the following queries given in Experiment Steps (5).

5. Experiment Steps:

Base Table:

Students

	student_id [PK] integer	name character varying (50)	dept_id integer
1	101	Aarav Sharma	10
2	102	Vihaan Iyer	10
3	103	Aditi Rao	20
4	104	Ananya Singh	20
5	105	Ishaan Gupta	30
6	106	Kavya Nair	30
7	107	Rahul Verma	40
8	108	Sanya Malhotra	40
9	109	Arjun Reddy	[null]
10	110	Siddharth Jain	[null]

Departments

	dept_id [PK] integer	dept_name character varying (50)
1	10	Computer Science
2	20	Electrical Engineering
3	30	Mechanical Engineering
4	40	Civil Engineering

Courses

	course_id [PK] character varying (10)	course_name character varying (100)
1	CS101	Artificial Intelligence
2	EE201	Power Systems
3	ME301	Thermodynamics
4	CE401	Structural Analysis
5	GE101	Environmental Studies
6	MA501	Advanced Calculus

Enrollments

	student_id [PK] integer	course_id [PK] character varying (10)
1	101	CS101
2	102	CS101
3	103	EE201
4	104	EE201
5	105	ME301
6	106	ME301
7	107	CE401
8	108	CE401
9	101	GE101
10	103	GE101
11	105	GE101
12	107	GE101

- a. Write queries to list students with their enrolled courses (INNER JOIN).

```
SELECT S.Name AS Student_Name, C.Course_Name FROM Students S
INNER JOIN Enrollments E ON S.Student_ID = E.Student_ID
INNER JOIN Courses C ON E.Course_ID = C.Course_ID
ORDER BY S.Name;
```

	student_name character varying (50)	course_name character varying (100)
1	Aarav Sharma	Environmental Studies
2	Aarav Sharma	Artificial Intelligence
3	Aditi Rao	Power Systems
4	Aditi Rao	Environmental Studies
5	Ananya Singh	Power Systems
6	Ishaan Gupta	Environmental Studies
7	Ishaan Gupta	Thermodynamics
8	Kavya Nair	Thermodynamics
9	Rahul Verma	Environmental Studies
10	Rahul Verma	Structural Analysis
11	Sanya Malhotra	Structural Analysis
12	Vihaan Iyer	Artificial Intelligence

- b. Find students not enrolled in any course (LEFT JOIN).

```
SELECT S.Name AS Unenrolled_Student FROM Students S
LEFT JOIN Enrollments E ON S.Student_ID = E.Student_ID
WHERE E.Course_ID IS NULL;
```

	unenrolled_student character varying (50)
1	Arjun Reddy
2	Siddharth Jain

c. Display all courses with or without enrolled students (RIGHT JOIN).

```
SELECT C.Course_Name, S.Name AS Student_Name
FROM Enrollments E JOIN Students S ON E.Student_ID = S.Student_ID
RIGHT JOIN Courses C ON E.Course_ID = C.Course_ID;
```

	course_name character varying (100)	student_name character varying (50)
1	Artificial Intelligence	Aarav Sharma
2	Artificial Intelligence	Vihaan Iyer
3	Power Systems	Aditi Rao
4	Power Systems	Ananya Singh
5	Thermodynamics	Ishaan Gupta
6	Thermodynamics	Kavya Nair
7	Structural Analysis	Rahul Verma
8	Structural Analysis	Sanya Malhotra
9	Environmental Studies	Aarav Sharma
10	Environmental Studies	Aditi Rao
11	Environmental Studies	Ishaan Gupta
12	Environmental Studies	Rahul Verma
13	Advanced Calculus	[null]

d. Show students with department info using SELF JOIN or multiple joins.

```
SELECT S.Name AS Student, D.Dept_Name, C.Course_Name
FROM Students S INNER JOIN Departments D ON S.Dept_ID = D.Dept_ID
INNER JOIN Enrollments E ON S.Student_ID = E.Student_ID
INNER JOIN Courses C ON E.Course_ID = C.Course_ID;
```

	student character varying (50)	dept_name character varying (50)	course_name character varying (100)
1	Aarav Sharma	Computer Science	Artificial Intelligence
2	Vihaan Iyer	Computer Science	Artificial Intelligence
3	Aditi Rao	Electrical Engineering	Power Systems
4	Ananya Singh	Electrical Engineering	Power Systems
5	Ishaan Gupta	Mechanical Engineering	Thermodynamics
6	Kavya Nair	Mechanical Engineering	Thermodynamics
7	Rahul Verma	Civil Engineering	Structural Analysis
8	Sanya Malhotra	Civil Engineering	Structural Analysis
9	Aarav Sharma	Computer Science	Environmental Studies
10	Aditi Rao	Electrical Engineering	Environmental Studies
11	Ishaan Gupta	Mechanical Engineering	Environmental Studies
12	Rahul Verma	Civil Engineering	Environmental Studies

e. Display all possible student-course combinations (CROSS JOIN).

```
SELECT S.Name AS Student, C.Course_Name AS Potential_Course
FROM Students S CROSS JOIN Courses C;
```

	student character varying (50) 	potential_course character varying (100) 
1	Aarav Sharma	Artificial Intelligence
2	Aarav Sharma	Power Systems
3	Aarav Sharma	Thermodynamics
4	Aarav Sharma	Structural Analysis
5	Aarav Sharma	Environmental Studies
6	Aarav Sharma	Advanced Calculus
....		
55	Siddharth Jain	Artificial Intelligence
56	Siddharth Jain	Power Systems
57	Siddharth Jain	Thermodynamics
58	Siddharth Jain	Structural Analysis
59	Siddharth Jain	Environmental Studies
60	Siddharth Jain	Advanced Calculus

6. Input/Output Details:

- The input for the experiment is the SQL queries mentioned in Experiment Steps (5).
- The output for each query is attached after the query itself.

7. Learning Outcome

- Understanding of the JOINS in SQL.
- Implementing referential integrity in SQL using foreign keys.
- Understanding of data relationships.